Test Series No.

1.13

MASTERY TEST SERIES

DURATION: 180 Minutes | **M. MARKS: 720**

DATE: 21/03/2026

SYLLABUS

Physics:	Full Syllabus * As per NTA
Chemistry:	Full Syllabus * As per NTA
Botany:	Full Syllabus * As per NTA
Zoology:	Full Syllabus * As per NTA

GENERAL INSTRUCTIONS:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration and the Test Booklet contains **180** multiple choice questions (four options with a single correct answer).
3. There is only one correct response for each question.
4. Each **correct** answer will give **4 Mark** while **1 Mark** will be **deducted** for a **wrong** answer. The maximum marks are 720.
5. No student is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone, any electronic device, etc. inside the examination room/hall.
6. On completion of the test, the candidate must hand over the Answer Sheet to the Invigilator on duty in the Room/Hall. However, the candidates are allowed to take away this Test Booklet with them.

OMR INSTRUCTIONS:

1. Use only blue/black dark ballpoint pen.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Student (In CAPITALS): _____

Roll Number: _____

OMR Bar Code Number: _____

Student's Signature: _____ **Invigilator's Signature:** _____

1. The dimensions formula of emf (electromotive force) is:

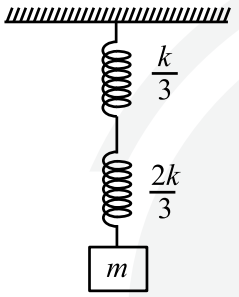
(1) $[M^0L^2T^{-3}A^{-1}]$

(2) $[ML^1T^{-3}A^{-1}]$

(3) $[ML^2T^{-2}A^{-1}]$

(4) $[ML^2T^{-3}A^{-1}]$

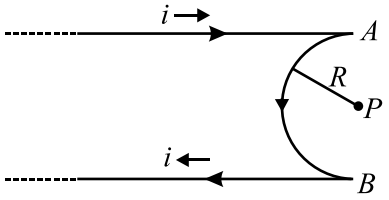
2. A block of mass m is suspended using two massless springs of spring constant $\frac{k}{3}$ and $\frac{2k}{3}$ respectively as shown in figure. If the block is slightly displaced vertically downward and released, then it executes SHM with time period.



(1) $2\pi\sqrt{\frac{m}{k}}$ (2) $2\pi\sqrt{\frac{m}{2k}}$

(3) $6\pi\sqrt{\frac{m}{2k}}$ (4) $6\pi\sqrt{\frac{m}{k}}$

3. A very long conducting wire is bent into two semi-infinite straight sections connected by a semi-circular arc as shown in figure. The net magnetic field at the center P of the arc for steady current i configuration is given by: (μ_0 is permeability of free space)



(1) $\frac{\mu_0 i}{4R}$ pointed out of the page

(2) $\frac{\mu_0 i}{4R}\left[1-\frac{2}{\pi}\right]$ pointed out of the page

(3) $\frac{\mu_0 i}{4R}\left[1-\frac{2}{\pi}\right]$ pointed into the page

(4) $\frac{\mu_0 i}{3R}\left[1+\frac{2}{\pi}\right]$ pointed into the page

4. If $\oint_s \vec{E}\cdot d\vec{S}=0$ over a surface, then:

(1) the magnitude of electric field on the surface is constant.

(2) all the charges must necessarily be inside the surface.

(3) the electric field inside the surface is necessarily uniform.

(4) the number of flux lines entering the surface must be equal to the number of flux lines leaving it.

5. A particle is moving three times as fast as an electron. The ratio of De-Broglie wavelength of the particle to that of an electron is 1.81×10^{-4}. Calculate the particle's mass and identify the particle.

(1) 1.275×10^{-26} kg, Neutron

(2) 1.576×10^{-27} kg, Proton

(3) 1.765×10^{-25} kg, Electron

(4) 1.675×10^{-27} kg, Neutron

6. An alternating voltage $V(t)=200\sin(100t)$ volts (where t is in second) is applied to a series LCR circuit containing a resistance of $10\,\Omega$, an inductance of $0.1\,\text{H}$, and a capacitance of $500\,\mu\text{F}$. In steady state, which of the following equations correctly represents the instantaneous voltage across the capacitor, $V_C(t)$?

(1) $200\sqrt{2}\sin\left(100t-\frac{\pi}{4}\right)\text{V}$

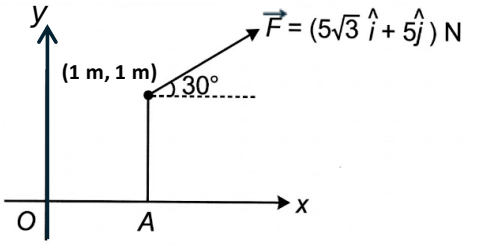
(2) $200\sqrt{2}\sin\left(100t+\frac{\pi}{4}\right)\text{V}$

(3) $100\sqrt{2}\sin\left(100t-\frac{3\pi}{4}\right)\text{V}$

(4) $100\sin\left(100t-\frac{\pi}{2}\right)\text{V}$

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7. Which of the following is the value of the torque of force F about origin O:



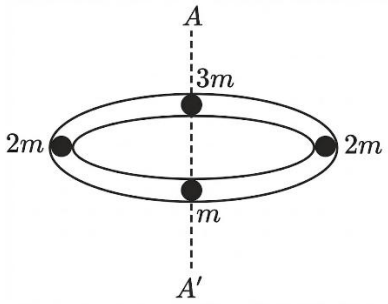
(1) $\vec{\tau} = 5(1-\sqrt{3})\hat{k}$ N-m

(2) $\vec{\tau} = 5(1-\sqrt{3})\hat{j}$ N-m

(3) $\vec{\tau} = 5(\sqrt{3}-1)\hat{i}$ N-m

(4) $\vec{\tau} = \sqrt{3}\hat{j}$ N-m

8. Four point masses are joined to a light circular frames as shown in the figure. The radius of gyration of this system about the axis AA' passing through the center of the circular frame and perpendicular to its plane would be: (where 'a' is the radius of the circle)



(1) $\frac{a}{\sqrt{2}}$ (2) $\frac{a}{2}$

(3) a (4) $2a$

9. Candela is the unit of:

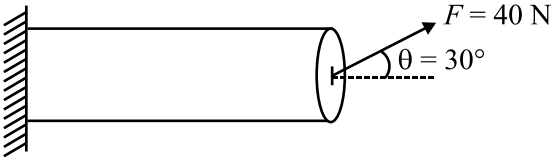
(1) Electric intensity

(2) Luminous intensity

(3) Sound intensity

(4) Magnetic intensity

10. A force of magnitude $F = 40$ N is applied at an angle $\theta = 30°$ with the horizontal at the free end of a horizontal cylindrical bar. If the cross-sectional area of the bar is $A = 2$ cm^2, the shear stress developed at this cross-section will be:



(1) 2.0×10^5 N/m^2 (2) 1.73×10^5 N/m^2

(3) 1.0×10^5 N/m^2 (4) 0.5×10^5 N/m^2

11. Which of the following is/are standard assumptions of an ideal Carnot engine?

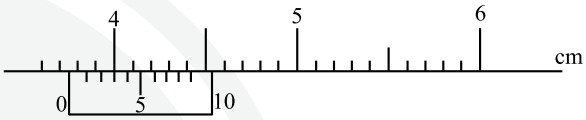
(1) All thermodynamic processes in the cycle are perfectly reversible and quasi-static.

(2) The piston moves with high velocity to ensure rapid expansion and compression.

(3) The heat source and sink have infinite thermal capacities, keeping their temperatures constant.

(4) Both (1) and (3)

12. Determine the **correct** measurement recorded by the Vernier Calliper as shown in the figure. Assume that the instrument has no zero error and its least count is 0.01 cm.



(1) 3.60 cm

(2) 3.64 cm

(3) 3.74 cm

(4) 4.04 cm

13. If C_p and C_v denote the specific heats of nitrogen per unit mass at constant pressure and constant volume respectively, then: (where R is the universal gas constant)

(1) $C_p - C_v = \frac{R}{28}$

(2) $C_p - C_v = \frac{R}{14}$

(3) $C_p - C_v = R$

(4) $C_p - C_v = 28R$

14. A charge of 4×10^{-9} C is uniformly distributed over the surface of a ring of radius 0.3 m. The intensity of electric field at a point on the axis of the ring at a distance of 0.4 m from the centre of the ring.

(1) 40.7 V/m

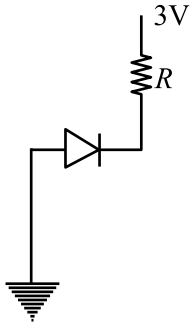
(2) 151.2 V/m

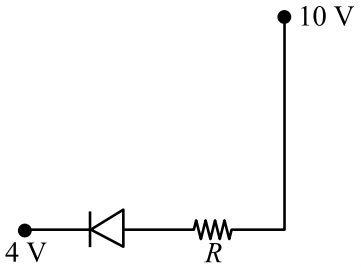
(3) 251.2 V/m

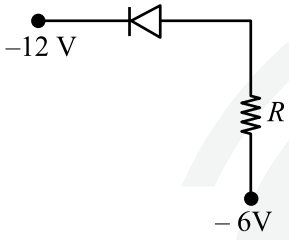
(4) 115.2 V/m

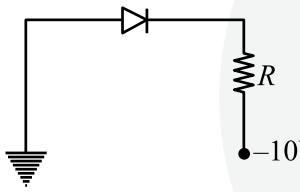
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15. Which one of the following diodes is reverse biased.

(1)


(2)


(3)


(4)


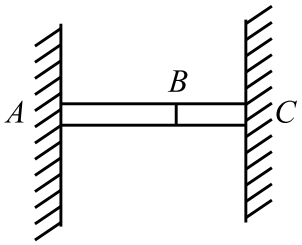
16. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:
Assertion (A): For a constant applied voltage, drift velocity of e^- in a metallic wire will decrease if temperature of wire is increased.
Reason (R): On increasing temperature conductivity of metallic wire decreases.
In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

17. A ship of mass 5×10^7 kg, initially at rest is pulled by a force of 2×10^4 N through a distance of 3 meters. Assuming that the resistance due to water is negligible, the speed of the ship is:

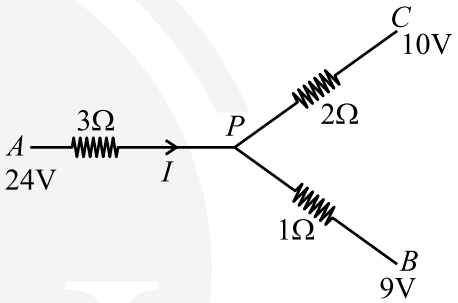
(1) 1 m/s (2) 0.5 m/s
(3) 0.1 m/s (4) 0.05 m/s

18. Two uniform rods AB and BC have Young's moduli 1.2×10^{11} N/m^2 and 1.5×10^{11} N/m^2 respectively. If the coefficient of linear expansion of AB is 1.5×10^{-5}/°C and both have equal area of cross section, then the coefficient of linear expansion of BC, for which there is no shift of the junction at all temperatures is:



(1) 1.5×10^{-5}/°C (2) 1.2×10^{-5}/°C
(3) 0.6×10^{-5}/°C (4) 0.75×10^{-5}/°C

19. In the circuit network shown in the figure, the value of current I is:



(1) 6 A (2) 2 A
(3) 4 A (4) 7 A

20. The momentum of a body is given by $p_x = 2t^2$ and $p_y = 3t$, where p is in kg m/s and t is in seconds.
Then choose the **incorrect** option:

(1) Momentum in x direction at t = 1s is 2 kg m/s
(2) Force in x direction at t = 1s is 4N
(3) Force in y direction is 3N
(4) Net force at t = 1s is 7N

21. Consider the following statements regarding electromagnetic waves:

(A) Both electric and magnetic fields are present in an electromagnetic wave.
(B) The energy density of the electric field is higher than that of the magnetic field.
(C) Both electric and magnetic fields contribute equally to the average energy density of the wave.

The False statement(s) is/are:

(1) (A), (B) only
(2) (B) only
(3) (A), (B) and (C)
(4) None of these

[Contd...

22. A bar magnet of magnetic moment $\vec{M}$ is placed in a uniform magnetic field $\vec{B}$. Let θ be the angle between the magnetic moment $\vec{M}$ and the magnetic field $\vec{B}$. Match the orientation of the magnet given in **List-I** with its corresponding potential energy (U) given in **List-II**, and select the **correct** option.

List-I (Orientation of the Magnet)		**List-II (Potential Energy, U)**	
(A)	Aligned parallel to the magnetic field ($\theta = 0°$)	(I)	$U = 0$
(B)	Aligned anti-parallel to the magnetic field ($\theta = 180°$)	(II)	$U = +MB$
(C)	Aligned perpendicular to the magnetic field ($\theta = 90°$)	(III)	$U = -\frac{MB}{2}$
(D)	Aligned at an angle of 60° to the magnetic field	(IV)	$U = -MB$

Choose the **correct** answer from the options given below:

(1) A-II, B-IV, C-I, D-III
(2) A-IV, B-II, C-I, D-III
(3) A-IV, B-II, C-III, D-I
(4) A-I, B-III, C-IV, D-II

23. Water is flowing continuously from a tap having an internal diameter 8×10^{-3} m. The water velocity as it leaves the tap is 0.4 m/s. The diameter of water stream at a distance of 0.4 m below the tap is close to: (g = 10 m/s^2)

(1) 2.9×10^{-3} m (2) 3.6×10^{-3} m
(3) 5.6×10^{-3} m (4) 1.1×10^{-3} m

24. The power loss in series LCR circuit connected to a constant voltage source will be minimum when:

(1) Resistance is high, inductance is high
(2) Resistance is high, inductance is low
(3) Resistance is low, inductance is low
(4) Resistance is low, inductance is high

25. A car of mass 1250 kg is moving at 30 m/s. Its engine delivers 30 kW of power while resistive force due to the surface is 750 N. What is the maximum acceleration the car can achieve at this instant.

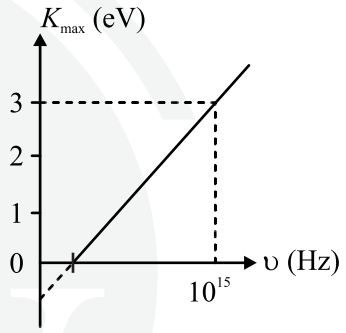
(1) $\frac{1}{3}$ m/s^2 (2) $\frac{1}{4}$ m/s^2
(3) $\frac{1}{5}$ m/s^2 (4) $\frac{1}{6}$ m/s^2

26. Choose the option containing **correct** statements.

A. A travelling wave transport energy but does not transfer momentum from one point to the another.
B. Mechanical waves are also known as elastic waves because these waves require medium for propagation.
C. With a decrease in humidity the speed of sound in air decreases.
D. If the temperature is kept fixed, the speed of sound is directly proportional to pressure.

(1) A, B and D only (2) B, C and D only
(3) B and C only (4) A and C only

27. If graph of kinetic energy of most energetic photoelectrons vs frequency for a metal used as cathode in photoelectric experiment is as shown in figure, then threshold frequency of light for the photoelectric emission from the metal is nearly:



(1) 3×10^{20} Hz (2) 3×10^{9} Hz
(3) 3×10^{7} Hz (4) 3×10^{14} Hz

28. If a body is taken up to a height of 1600 km from the earth's surface, then the percentage loss of gravitational force acting on that body will be: (Radius of earth R_e = 6400 km).

(1) 50% (2) 36%
(3) 25% (4) 10%

29. A circular loop of radius r is placed in a region where magnetic field perpendicular to plane of loop increases with respect to time t as $B(t) = (at)$ where a is constant, then magnitude of induced emf in coil is:

(1) $\pi r^2 a$ (2) $3\pi r^2 a$
(3) $2\pi r^2 a$ (4) $4\pi r^2 a$

30. The colour of light emitted by LED depends on:

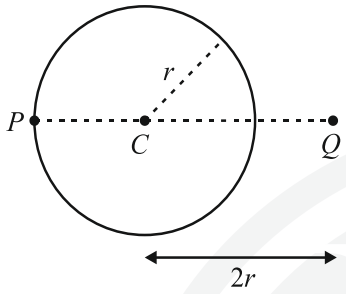
(1) its reverse bias.
(2) the amount of reverse current.
(3) type of semiconductor material used.
(4) both (2) and (3)

[Contd...

31. A body of length 1 m having cross-sectional area 0.75 m^2 conducts heat at the rate of 9000 J/s. The temperature difference between two ends of conductor if $K = 200\ W\ m^{-1}K^{-1}$ is:

(1) 20°C (2) 60°C
(3) 80°C (4) 100°C

32. A charge Q is placed in front of an isolated, uncharged conducting sphere at a distance $2r$ from center of sphere as shown in the figure. Potential at point C at the centre of the sphere due to induced charges is:



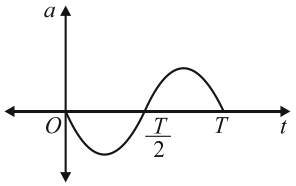
(1) $\frac{kQ}{6r}$ (2) $\frac{kQ}{2r}$

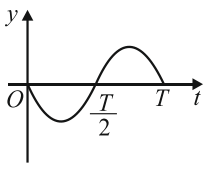
(3) $\frac{kQ}{3r}$ (4) zero

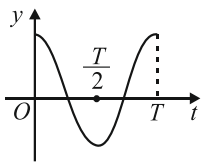
33. If an equiconvex lens of focal length $2f$ is cut into two identical parts by plane along its optical axis (principal axis), then the focal length of each part will be;

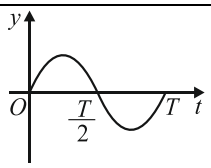
(1) f and the intensity of image will be increased
(2) $4f$ and the intensity of image will be decreased
(3) $2f$ and the intensity of image will be increased
(4) $2f$ and the intensity of image will be decreased

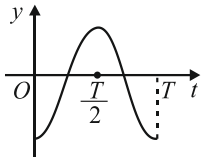
34. A body is executing simple harmonic motion along y axis whose acceleration a versus time t graph is shown in figure. Its corresponding position y versus time t graph is;



(1)


(2)


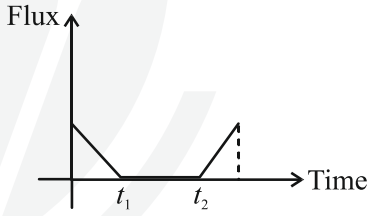
(3)


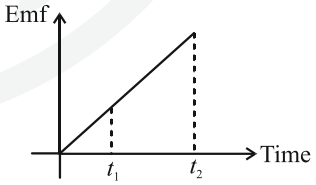
(4)


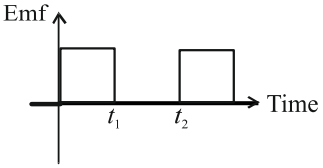
35. A ball is thrown vertically upward with some speed from the ground. The ball passes a point at h height above the ground at time 4 s and 12 s respectively. The speed with which ball is thrown up from the ground is ($g = 10\ m/s^2$)

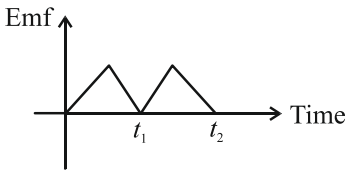
(1) 60 m/s
(2) 75 m/s
(3) 80 m/s
(4) 82.5 m/s

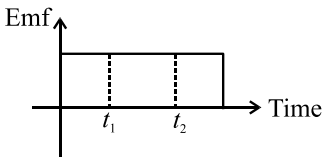
36. The variation of magnetic flux with time is shown. The variation of magnitude of induced emf with time is shown as:



(1)


(2)


(3)


(4)


[Contd...

37. If an ideal monoatomic gas is heated at constant pressure then percentage of total heat given, that will be used to do external work is:

(1) 40% (2) 60%

(3) 25% (4) 20%

38. A particle of mass 2 kg located at the position $(\hat{i}+\hat{j})$ m has a velocity $2(+\hat{i}-\hat{j}+\hat{k})$ m/s. Its angular momentum about z-axis in kg-m^2/s is:

(1) zero (2) $+8\hat{k}$

(3) $12\hat{k}$ (4) $-8\hat{k}$

39. The velocity of electron in second Bohr orbit of H-atom is

(1) 43.8×10^5 m/s

(2) 10.9×10^5 m/s

(3) 2.18×10^6 m/s

(4) 21.0×10^5 m/s

40. A nuclear fission is given below

$A^{240} \rightarrow B^{100} + C^{140} + Q$ (energy)

Let the binding energy per nucleon of nucleus A, B and C be 7.6 MeV, 8.1 MeV and 8.1 MeV respectively. Value of Q is: (Approximately)

(1) 20 MeV (2) 220 MeV

(3) 120 MeV (4) 240 MeV

41. What happens when the width of the slit aperture is increased in an experiment of single slit diffraction experiment?

(1) spread of diffraction region is increased.

(2) spread of diffraction region remain same.

(3) spread of diffraction region will be decreased and midband becomes narrow.

(4) None of these

42. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): Shape of wavefront in case of Fraunhofer diffraction is nearly planar.

Reason (R): In case of Fraunhofer diffraction, distance between source of light and screen is effectively infinite.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true but R is NOT the correct explanation of A.

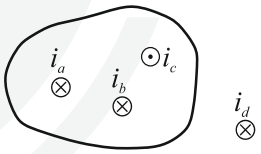
43. A convex lens is made up of three different materials as shown in the figure. For a point object placed on its axis, the number of images formed are;



(1) 1 (2) 3

(3) 4 (4) 5

44. If $\vec{B}_a, \vec{B}_b, \vec{B}_c$ and $\vec{B}_d$ are magnetic fields produced due to steady currents i_a, i_b, i_c and i_d respectively as shown in figure then $\vec{B}$ in the Ampere's circuital law $\oint \vec{B}.d\vec{l} = \mu_0 i_{\text{inside}}$ is:



(1) $\vec{B}_a + \vec{B}_b + \vec{B}_c + \vec{B}_d$

(2) $\vec{B}_a + \vec{B}_b + \vec{B}_c$

(3) $\vec{B}_c - \vec{B}_a - \vec{B}_b$

(4) $-\vec{B}_d$

45. A parallel plate capacitor is charged by connecting it to a battery. A dielectric slab is inserted into the space between the plates with the battery still connected.

A. The charge on the plates increases.

B. The energy stored in the capacitor increases.

C. Work is done by the battery as the slab is inserted.

Choose the option containing **correct** statement/s.

(1) A and B (2) A and C

(3) Only A (4) A, B and C

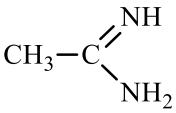
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46. Favourable conditions for dissociation of NH_4HS are:

$NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g),\ \Delta H > 0$

(1) High temperature and low pressure
(2) High temperature and high pressure
(3) low temperature and low pressure
(4) low temperature and high pressure

47. The **correct** order of basicities of the following compounds is:

(I) 

(II) $CH_3CH_2NH_2$
(III) $(C_2H_5)_2NH$
(IV) CH_3CONH_2
(1) I > III > II > IV
(2) III > IV > I > II
(3) I > II > III > IV
(4) II > III > IV > I

48. 4 L of an ideal gas is allowed to expand isothermally into vacuum until the total volume is 8 L. The amount of Heat absorbed in this expansion is:
(1) 2 L atm.
(2) zero
(3) 6 L atm
(4) 10 L atm

49. Given below are two statements:
Statement I: Average oxidation state of carbon in C_3O_2 is $+\frac{4}{3}$.
Statement II: Fe_3O_4 is a mixed oxide
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

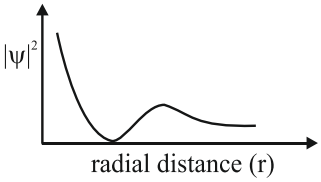
50. The set of elements that differ in mutual relationship from those of the other sets is:
(1) Li and Na
(2) Be and Al
(3) Li and Mg
(4) B and Si

51. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: Kjeldahl method is not applicable to estimate nitrogen in nitrobenzene.
Reason R: Nitrogen present in nitrobenzene does not change to $(NH_4)_2SO_4$ during the process.
In the light of above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

52. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:
Assertion A: In O_2F_2, oxidation state of oxygen is positive.
Reason R: Fluorine is more electronegative than oxygen.
In the light of above statements, choose the **correct** answer from the options given below:
(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

53. Given below are two statements:
Statement I: Dalton's law could explain the laws of gaseous volumes.
Statement II: Precision refers to the closeness of various measurements for the same quantity.
In the light of the above statements, choose the *most appropriate* answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

54. The given graph represents:



(1) 1s-orbital (2) 2p-orbital
(3) 2s-orbital (4) 3p-orbital

[Contd...

55. The shortcomings of valence bond theory for coordination compounds are:

(A) It does not explain the colour exhibited by coordination compounds.

(B) It does not distinguish between weak and strong ligands.

(C) It does not give quantitative interpretation of magnetic data.

(1) A and B only

(2) B and C only

(3) A and C only

(4) A, B and C

56. Which among the following elements show greatest number of oxidation states?

(1) Mn

(2) Fe

(3) V

(4) Ni

57. Match **List-I** with **List-II**.

List-I (Unit of rate constant)		**List-II (Order of reaction)**	
(A)	$mol^{-1}\ L\ s^{-1}$	(I)	One
(B)	s^{-1}	(II)	Two
(C)	$mol\ L^{-1}\ s^{-1}$	(III)	Three
(D)	$mol^{-2}\ L^{2}\ s^{-1}$	(IV)	Zero

Choose the **correct** answer from the options given below:

(1) A-III; B-I; C-IV; D-II

(2) A-II; B-III; C-I; D-IV

(3) A-I; B-III; C-IV; D-II

(4) A-II; B-I; C-IV; D-III

58. The **correct** formula of gypsum is:

(1) $CaSO_4$

(2) $CaSO_4 \cdot \frac{1}{2}H_2O$

(3) $CaSO_4 \cdot H_2O$

(4) $CaSO_4 \cdot 2H_2O$

59. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:

Assertion A: Vapour pressure of water increases on increasing the temperature.

Reason R: Vapour pressure is not a colligative property.

In the light of above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true and R is NOT the correct explanation of A.

60. Match **List-I** with **List-II**.

List-I (Process)		**List-II (Reagent used)**	
(A)	Aromatization of n-heptane	(I)	Cr_2O_3, 773K, 10-20 atm
(B)	Isomerisation of n-butane	(II)	Red hot iron tube, 873K
(C)	Polymerisation of acetylene	(III)	Zn, Δ
(D)	Dehalogenation of ethylene dichloride	(IV)	Anhyd. $AlCl_3$/HCl

Choose the **correct** answer from the options given below:

(1) A-III; B-I; C-IV; D-II

(2) A-II; B-III; C-I; D-IV

(3) A-I; B-IV; C-II; D-III

(4) A-II; B-I; C-IV; D-III

61. Match **List-I** with **List-II**.

List-I (Solution)		**List-II (pH)**	
(A)	10^{-3} M HNO_3	(I)	1.7
(B)	0.01 M H_2SO_4	(II)	12
(C)	10^{-7} M NaOH	(III)	3
(D)	10^{-2} M NaOH	(IV)	7.3

Choose the **correct** answer from the options given below:

(1) A-II; B-III; C-I; D-IV

(2) A-III; B-I; C-IV; D-II

(3) A-I; B-IV; C-II; D-III

(4) A-II; B-I; C-IV; D-III

62. The electrolysis product at anode during electrolysis of concentrated sulphuric acid is:

(1) $O_2(g)$

(2) $SO_3(g)$

(3) $SO_2(g)$

(4) $S_2O_8^{2-}(aq)$

[Contd...

63. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Canonical forms of CO_2	(I)	1.5
(B)	S–O bond order in SO_4^{2-}	(II)	1
(C)	Lone pair on central atom in XeO_3	(III)	1.33
(D)	C–O bond order in CO_3^{2-}	(IV)	3

Choose the **correct** answer from the options given below:

(1) A-II; B-III; C-I; D-IV
(2) A-III; B-I; C-IV; D-II
(3) A-I; B-IV; C-II; D-III
(4) A-IV; B-I; C-II; D-III

64. Given below are two statements:

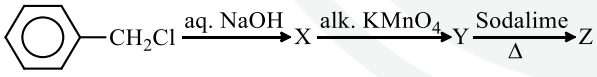
Statement I: Fehling solution contains tartarate salt.

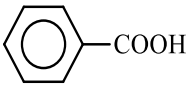
Statement II: Reagent used in etard reaction is chromyl chloride.

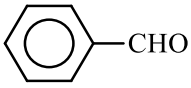
In the light of the above statements, choose the *most appropriate* answer from the options given below:

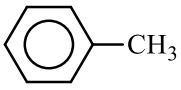
(1) Both Statement I and Statement II are correct.
(2) Statement I is correct but Statement II is incorrect.
(3) Statement I is incorrect but Statement II is correct.
(4) Both Statement I and Statement II are incorrect.

65. The major product (Z) in the following reaction sequence is:



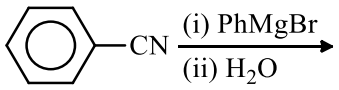
(1) 

(2) 

(3) 

(4) 

66. Major product of the following reaction is:

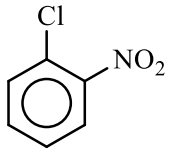


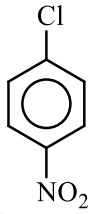
(1) Mesityl oxide
(2) Acetophenone
(3) Benzophenone
(4) Benzoquinone

67. The alcohol which react at fastest rate with conc. HCl and $ZnCl_2$ is:

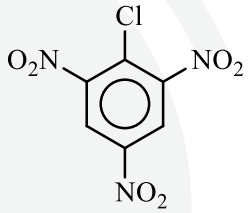
(1) Tert-butanol (2) n-Butanol
(3) Phenol (4) sec-Butanol

68. The compound which react fastest with NaOH is:

(1) 

(2) 

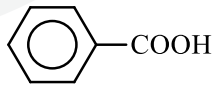
(3) 

(4) 

69. Among the following the most reactive compound towards S_N1 reaction is:

(1) $C_6H_5CH_2Cl$ (2) C_6H_5Cl
(3) $C_6H_5CH(Cl)CH_3$ (4) $(C_6H_5)_3CCl$

70. Strongest acid among the following is:

(1) O_2NCH_2–COOH
(2) HCOOH
(3) 

(4) CCl_3COOH

71. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:

Assertion A: Carbon can show property of catenation.

Reason R: C–C bonds are very strong.

In the light of above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

[Contd...

72. In nitration and sulphonation of benzene the electrophiles are:

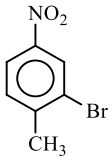
(1) NO_2^+ and SO_3 respectively

(2) NO_2^+ and SO_3H^+ respectively

(3) NO_2 and SO_2 respectively

(4) NO^+ and SO_3^+ respectively

73. IUPAC name of given compound is:



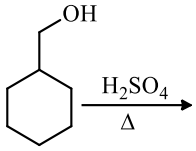
(1) 4-Bromo-5-bromo-1-nitrobenzene

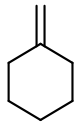
(2) 1-Bromo-2-methyl-5-nitrobenzene.

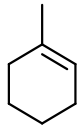
(3) 2-Bromo-1-methyl-4-nitrobenzene

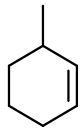
(4) 2-Bromo-4-nitro-1-methylbenzene

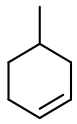
74. Major product of the following reaction is:



(1) 

(2) 

(3) 

(4) 

75. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:

Assertion A: Ethane forms only one monosubstituted derivative.

Reason R: All the hydrogens in ethane are equivalent.

In the light of above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true and R is NOT the correct explanation of A.

76. Given below are two statements:

Statement I: Keratin is an example of fibrous protein.

Statement II: Globular proteins are generally insoluble in water.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Both Statement I and Statement II are correct.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct.

(4) Both Statement I and Statement II are incorrect.

77. The rate constant for a first order reaction is 0.1 s^{-1}. The time required for 200 g of the reactant to reduce to 2 g is:

(1) 23.03 s (2) 34.54 s

(3) 53.18 s (4) 46.06 s

78. Assuming Hund's rule is violated, the bond order and magnetic nature of B_2 molecule is:

(1) 1 and paramagnetic

(2) 0 and diamagnetic

(3) 2 and diamagnetic

(4) 1 and diamagnetic

79. Volume of O_2 gas required at STP for the complete combustion of 7.8g of benzene is:

(1) 16.8 L (2) 8.4 L

(3) 12.6 L (4) 25.2 L

80. Non-redox reaction among the following is:

(1) $Zn + 2HCl \rightarrow ZnCl_2 + H_2$

(2) $2F_2 + 2H_2O \rightarrow 4HF + O_2$

(3) $NaH + H_2O \rightarrow NaOH + H_2$

(4) $Ca(OH)_2 + H_2SO_4 \rightarrow CaSO_4 + 2H_2O$

81. Match **List-I** with **List-II**.

List-I (Element)		List-II (IUPAC official name)	
(A)	Unu	(I)	Bohrium
(B)	Unq	(II)	Lawrencium
(C)	Unt	(III)	Rutherfordium
(D)	Uns	(IV)	Mendelevium

Choose the **correct** answer from the options given below:

(1) A-II; B-III; C-I; D-IV

(2) A-IV; B-III; C-II; D-I

(3) A-I; B-IV; C-II; D-III

(4) A-II; B-I; C-IV; D-III

[Contd...

82. The ΔH° for the given reaction is:

$H_2(g) + Br_2(g) \rightarrow 2HBr(g)$

(Given: Bond energies of H_2, Br_2 and HBr are 433, 192 and 364 kJ mol^{-1} respectively)

(1) –91 kJ mol^{-1}

(2) –312 kJ mol^{-1}

(3) –215 kJ mol^{-1}

(4) –103 kJ mol^{-1}

83. Given below are two statements: one is labelled as Assertion A and other is labelled as Reason R:

Assertion A: $KMnO_4$ titrations in presence of HCl are unsatisfactory.

Reason R: $KMnO_4$ oxidises HCl into Cl_2.

In the light of above statements, choose the **correct** answer from the options given below:

(1) A is true but R is false.

(2) A is false but R is true.

(3) Both A and R are true and R is the correct explanation of A.

(4) Both A and R are true and R is NOT the correct explanation of A.

84. Given below are two statements:

Statement I: $[Co(NH_3)_4Cl_2]^+$ is a homoleptic coordination entity.

Statement II: Hardness of water is estimated by simple titration with Na_2 EDTA.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

(1) Both Statement I and Statement II are correct.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct.

(4) Both Statement I and Statement II are incorrect.

85. The energy required to remove an electron from ground state of He^+ ion is:

(1) 13.6 eV

(2) 54.4 eV

(3) 3.4 eV

(4) 122.4 eV

86. The value of van't Hoff factor (i) for 80% dimerisation of ethanoic acid in benzene is:

(1) 0.8

(2) 0.4

(3) 0.2

(4) 0.6

87. The resistivity of 0.1 M solution of an CH_3COOH is 1×10^{-4} ohm cm. Its molar conductance (in ohm^{-1} cm^2 mol^{-1}) will be:

(1) 1×10^8

(2) 1×10^7

(3) 1×10^6

(4) 1×10^5

88. Borax bead test gives characteristic colours of:

(1) alkali metals

(2) alkaline earth metals

(3) transition metals

(4) lanthanoid metals

89. Which among the following elements has lowest electronegativity?

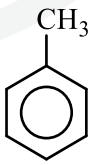
(1) Al

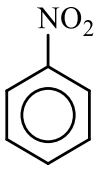
(2) Ga

(3) In

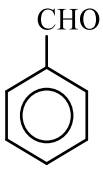
(4) Tl

90. Which of the following is most reactive towards electrophilic substitution reaction?

(1) 

(2) 

(3) 

(4) 

[Contd...

91. Watson and Crick were able to propose the complementary double-helical configuration of the DNA in their ____ effort.
(1) first (2) second
(3) third (4) fourth

92. Select the **correct** sequence of phases in the cell cycle.
(1) $G_1 \rightarrow S \rightarrow G_2 \rightarrow M$
(2) $S \rightarrow G_2 \rightarrow G_1 \rightarrow M$
(3) $M \rightarrow G_1 \rightarrow G_2 \rightarrow S$
(4) $G_2 \rightarrow G_1 \rightarrow S \rightarrow M$

93. Given below are two statements:
Statement I: The five-kingdom classification system did not distinguish between the eukaryotes and prokaryotes.
Statement II: *Chlamydomonas* and *Spirogyra* were placed together under algae in two-kingdom classification systems.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

94. The stromal lamellae in chloroplast lacks;
(1) Light harvesting complex.
(2) Reaction centre.
(3) NADP reductase.
(4) Photosystem I.

95. Consider the following statements regarding the regions of the root:
A. The root cap protects the tender apex of the root.
B. The cells of the region of maturation are very small and thin-walled.
C. The region of maturation is present proximal to the region of elongation.
D. Root hairs are formed from some of the epidermal cells in the region of meristematic activity.
E. The region of elongation is responsible for the growth of the root in length.
The **incorrect** statements are;
(1) B and D (2) D and E
(3) A and E (4) A and C

96. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): A man and woman homozygous for blood group A and B, respectively, cannot have a child with blood group O.
Reason (R): For blood group O, the child must inherit one *i* allele from each of the parents.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

97. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Casparian strips	(I)	Dicot stem
(B)	Starch sheath	(II)	Dicot root
(C)	Bulliform cells	(III)	Grass leaf
(D)	Sclerenchymatous hypodermis	(IV)	Monocot stem

Choose the *most appropriate* answer from the options given below.
(1) A-IV, B-III, C-II, D-I
(2) A-II, B-I, C-III, D-IV
(3) A-I, B-III, C-II, D-IV
(4) A-III, B-IV, C-I, D-II

98. Read the following statements regarding growth.
A. It is a defining feature of living organisms.
B. It is said to be indeterminate in plants.
C. Growth of a watermelon cell is measured as increase in surface area.
D. Non-living objects grow by deposition of material on external surface.
E. In maize root apical meristem, growth is expressed in terms of increase in cell number.
How many of the given statements are **correct**?
(1) Zero (2) Four
(3) Five (4) Three

99. Which of the following cells in a mature pollen grain is responsible for giving rise to the two male gametes?
(1) Vegetative cell
(2) Generative cell
(3) Tapetal cell
(4) Intine cell

100. Given below are two statements:
Statement I: Primary treatment of sewage involves biological decomposition of organic matter.
Statement II: BOD refers to the amount of oxygen that would be consumed if all the organic matter in one liter of water were oxidised by bacteria.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

101. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): The Law of Segregation is based on the fact that the alleles do not show any blending.
Reason (R): In Mendel's experiments, both the characters were recovered as such in the F_2 generation though one of these was not seen at the F_1 stage.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

102. Identify the correct statements.
A. The population size of Siberian cranes in Bharatpur wetland can be as low as <10.
B. The size of a population does not depend on any factor.
C. The population size of *Chlamydomonas* in a pond can go into millions.
D. The tiger census in our national parks and tiger reserves is often based on pug marks and fecal pellets.
E. The number of fish caught per trap is good enough measure of its total population density in the lake.
Choose the **correct** answer from the options given below:
(1) A, B, C only (2) C, D, E only
(3) A, C, D, E only (4) B, C, D only

103. Which among the following is not a producer in the pond ecosystem?
(1) Submerged plants
(2) Zooplanktons
(3) Floating plants
(4) Phytoplanktons

104. Which of the following statements regarding biodiversity patterns are **correct**?
A. Species diversity increases as we move away from the equator towards the poles.
B. Tropics harbor more species than temperate or polar areas.
C. Colombia has nearly 1,400 species of birds.
D. Amazonian rain forest is home to more than 40,000 species of plants.
E. The species-area relationship for a wide variety of taxa turns out to be a rectangular hyperbola.
Choose the **correct** answer from the options given below:
(1) A, B and C only
(2) B, C, D and E only
(3) A, C and E only
(4) B, D and E only

105. The cell theory as hypothesised by Schleiden and Schwann was modified by;
(1) Antonie Von Leeuwenhoek.
(2) Camillo Golgi.
(3) Rudolf Virchow.
(4) Robert Brown.

106. Identify the **incorrectly** matched pair.
(1) *Chlorella* - Rich in proteins
(2) *Sphagnum* - Provides peat
(3) *Cycas* - Coralloid roots
(4) *Marchantia* - Possess xylem and phloem

107. Match **List-I** with **List-II**.

List-I (Organism)		List-II (Order)	
(A)	Man	(I)	Diptera
(B)	Housefly	(II)	Sapindales
(C)	Mango	(III)	Poales
(D)	Wheat	(IV)	Primata

Choose the *most appropriate* answer from the options given below.
(1) A-IV, B-I, C-II, D-III
(2) A-IV, B-II, C-I, D-III
(3) A-I, B-III, C-IV, D-II
(4) A-III, B-IV, C-I, D-II

108. If a C_3 plant is grown in a greenhouse with a CO_2 concentration of 450 μlL^{-1}, what is the most likely effect on its rate of photosynthesis compared to ambient conditions?
(1) The rate will decrease due to CO_2 toxicity.
(2) The rate will show no change C_3 plants saturate at 360 μlL^{-1} of CO_2.
(3) The rate will increase as saturation for C_3 is seen only beyond 450 μlL^{-1} of CO_2.
(4) The rate will fluctuate depending on O_2 levels only.

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109. Regarding glycolysis, which of the following statements are correct?

A. It occurs in the cytoplasm of the cell.
B. Glucose undergoes complete oxidation to form two molecules of pyruvic acid.
C. ATP is utilised at two steps.
D. NADH + H^+ is formed during the conversion of PGAL to BPGA.
E. It is the only process in respiration for anaerobic organisms.

Choose the **correct** answer from the options given below:

(1) A, B, C and D (2) A, C, D and E
(3) B, C, D and E (4) A, B, D and E

110. Match **List-I** with **List-II**.

List-I		List-II	
(A)	VNTR	(I)	0.1 to 20 kb
(B)	Nucleosome	(II)	200 bp
(C)	Haploid content of human DNA	(III)	3.3×10^9 bp
(D)	*E. coli* DNA	(IV)	4.6×10^6 bp

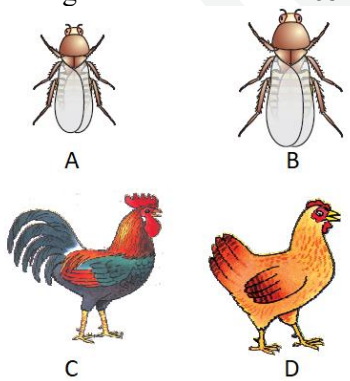
Choose the *most appropriate* answer from the options given below.

(1) A-I, B-II, C-III, D-IV
(2) A-II, B-I, C-III, D-IV
(3) A-IV, B-III, C-II, D-I
(4) A-III, B-IV, C-I, D-II

111. Cuticle is absent in;

(1) dorsiventral leaf.
(2) roots.
(3) isobilateral leaf.
(4) stem.

112. Identify the sex chromosomes of the organisms in the given diagram and choose the **correct** option.



(1) A - XY; B - XX; C - ZW; D – ZZ
(2) A - XX; B - XY; C - ZW; D – ZZ
(3) A - XY; B - XX; C - ZZ; D – ZW
(4) A - XX; B - XY; C - ZZ; D – ZW

113. Given below are two statements:

Statement I: Meiosis ensures the production of haploid phase in the life cycle of sexually reproducing organisms.

Statement II: Meiosis involves two sequential cycles of nuclear and cell division and two cycles of DNA replication.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

114. Identify the **correctly** matched pair.

(1) Tobacco Mosaic Virus - Double stranded DNA
(2) Bacteriophages - Single stranded RNA
(3) Viruses infecting animals - Either single or double stranded RNA or double stranded DNA
(4) Capsid - Made of carbohydrate subunits

115. How does self-incompatibility prevent inbreeding in flowering plants?

(1) By placing anther and stigma at different positions.
(2) By inhibiting pollen germination or pollen tube growth in the pistil.
(3) By producing unisexual flowers on the same plant.
(4) By non-synchronised pollen release and stigma receptivity.

116. In pteridophytes, upon germination the spores give rise to;

(1) protonema.
(2) prothallus.
(3) secondary protonema.
(4) strobilus.

117. Match **List-I** with **List-II** w.r.t symbols of floral formula.

List-I		List-II	
(A)	⊕	(I)	Calyx
(B)	%	(II)	Corolla
(C)	K	(III)	Actinomorphic flower
(D)	C	(IV)	Zygomorphic flower

Choose the *most appropriate* answer from the options given below.

(1) A-I, B-II, C-III, D-IV
(2) A-I, B-III, C-II, D-IV
(3) A-II, B-I, C-III, D-IV
(4) A-III, B-IV, C-I, D-II

118. If 1 glucose molecule undergoes complete aerobic respiration, how many ATP will be synthesised only by oxidative phosphorylation of all NADH + H^+ produced during the process?

(1) 38 (2) 30
(3) 2 (4) 3

[Contd...

119. _____ has been called The Darwin of the 20th century.
(1) Ernst Mayr (2) J. A. Bassham
(3) Katherine Esau (4) W. H. Pearsall

120. A bioactive molecule used as blood-cholesterol lowering agent is;
(1) Cyclosporin A. (2) Streptokinase.
(3) Statin. (4) Lipase.

121. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A): Mitochondria are called 'power houses' of the cell.
Reason (R): Mitochondria divide by fission.
In the light of the above statements, choose the *most appropriate* answer from the options given below.
(1) Both (A) and (R) are true and (R) is the correct explanation of (A).
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(3) (A) is true but (R) is false.
(4) (A) is false but (R) is true.

122. The organisms responsible for the production of biogas from the dung of ruminant animals;
(1) are obligate aerobes.
(2) lack cell wall.
(3) are also used as biofertilisers in paddy fields.
(4) are classified as archaebacteria.

123. Differentiation of xylem is controlled by;
(1) auxins. (2) ethylene.
(3) abscisic acid. (4) cytokinins.

124. The companion cells of phloem;
(1) lack nucleus.
(2) are not present in angiosperms.
(3) control the function of sieve tubes.
(4) are specialised collenchymatous cells.

125. In Photosystem I, the reaction centre chlorophyll *a* has an absorption peak at;
(1) 680 nm. (2) 700 nm.
(3) 450 nm. (4) 500 nm.

126. Arrange the following layers of a microsporangium in correct sequence from outside to inside.
A. Tapetum B. Middle layers
C. Epidermis D. Endothecium
Choose the **correct** option.
(1) A → B → C → D
(2) D → C → A → B
(3) B → A → D → C
(4) C → D → B → A

127. The probability of a son born to a woman who is a carrier of colour blindness and married to a normal man, is;
(1) 100%. (2) 0%.
(3) 25%. (4) 50%.

128. The synthesis of tRNA occurs by the process of;
(1) replication. (2) translation.
(3) transcription. (4) phosphorylation.

129. The type of population interaction between *Chathamalus* and *Balanus* on the rocky sea coasts of Scotland is;
(1) Mutualism. (2) Parasitism.
(3) Competition. (4) Amensalism.

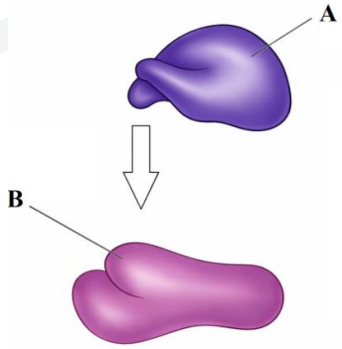
130. The top strata in a forest is occupied by;
(1) trees. (2) shrubs.
(3) herbs. (4) grasses.

131. Clearing of Amazon rainforests for cultivation of soyabeans is an example of;
(1) Over-exploitation.
(2) Alien species invasions.
(3) Habitat loss and fragmentation.
(4) Co-extinctions.

132. The codon for Glutamate is;
(1) GUG. (2) UAG.
(3) AUG. (4) GAG.

133. The DNA-dependent DNA polymerase can catalyse polymerisation only in the direction, that is;
(1) 3'→5' (2) 5'→3'
(3) 3'→3' (4) 5'→5'

134. Given below are the two subunits of a ribosome. Choose the **correct** statement about it.



(1) When 'A' encounters an mRNA, the process of translation begins.
(2) In prokaryotes, 'A' is 60S.
(3) In eukaryotes, 'B' is 30S.
(4) 'B' is the smaller sub unit.

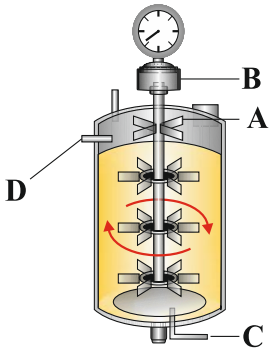
135. Which of the following is similar to amylopectin and glycogen in structure?
(1) Floridean starch (2) Laminarin
(3) Chitin (4) Mannitol

136. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: In echinoderms, the water vascular system plays a critical role in locomotion, capture and transport of food, and respiration.
Reason R: The water vascular system in echinoderms is different from the water canal system of Porifera.
In the light of the above statements, choose the **correct** answer from the options given below:
(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

137. Read the following statements and choose the option with **correct** statements.
A. A protein is imagined as a line, the left end represented by N-terminal and the right end by C-terminal.
B. The tertiary structure is absolutely necessary for the many biological activities of proteins.
C. In the presence of competitive inhibitor, the k_m decreases significantly.
D. Co-enzymes are organic compounds that are tightly and permanently bound to the apoenzyme.
E. Carbonic anhydrase accelerates the reaction between CO_2 and H_2O by about 10 million times.
(1) A, B and E
(2) A, B and C
(3) B, C, D and E
(4) A, C and D

138. Identify the **incorrect** statement regarding the anatomical features of *Periplaneta americana*:
(1) A ring of 6-8 blind tubules called gastric caeca is present at the junction of foregut and midgut.
(2) The hindgut is broader than the midgut.
(3) Excretion is performed by 100-150 yellow coloured thin filamentous tubules lined by glandular and ciliated cells.
(4) Blood from sinuses enters the elongated muscular heart through ostia and is pumped posteriorly to sinuses again.

139. Which of the following represents foam breaker?



(1) B (2) A
(3) C (4) D

140. Match **List-I** with **List-II.**

List-I		List-II	
(A)	P-wave	(I)	Comparatively thin in the veins
(B)	Tunica intima	(II)	Depolarisation of the atria
(C)	T-wave	(III)	Return of ventricles from excited to normal state
(D)	Tunica media	(IV)	An inner lining of squamous endothelium

Choose the **correct** answer from the options given below:
(1) A-II, B-IV, C-III, D-I
(2) A-II, B-I, C-III, D-IV
(3) A-IV, B-II, C-I, D-III
(4) A-II, B-IV, C-I, D-III

141. Which of the following will **most** likely stimulate the release of Aldosterone from the adrenal cortex?
(1) An excessive increase in body fluid volume
(2) An increase in blood flow to the atria
(3) A fall in glomerular blood flow
(4) A sudden drop in the levels of angiotensin II.

142. During muscle contraction, calcium ions bind directly to which protein to remove the masking of active sites on the actin filament?
(1) Tropomyosin (2) Troponin
(3) Myosin (4) Actin

143. During the resting state of a neuron, if the Na^+-k^+ pump is blocked by a metabolic inhibitor, what will be the **most** immediate consequence?
(1) The membrane will still remain in polarised state.
(2) The concentration gradients of Na^+ and K^+ across the membrane will gradually be lost.
(3) The axoplasm will instantly become highly negatively charged.
(4) Voltage-gated calcium channels at the synaptic knob will permanently open.

[Contd...

144. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Glucagon	(I)	Stimulates lipolysis, and proteolysis
(B)	Cortisol	(II)	A hypocalcemic hormone
(C)	Aldosterone	(III)	Mineralocorticoid
(D)	Thyrocalcitonin	(IV)	Acts as antagonist to insulin

Choose the **correct** answer from the options given below:

(1) A-I, B-IV, C-III, D-II
(2) A-IV, B-I, C-II, D-III
(3) A-IV, B-I, C-III, D-II
(4) A-III, B-II, C-I, D-IV

145. Consider the following statements:

Statement I: In human females, the process of oogenesis is initiated during the embryonic development stage.

Statement II: In human males, spermatogenesis begins at puberty due to a significant increase in the secretion of GnRH.

In the light of the above statements, choose the most appropriate answer from the options given below

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect.
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

146. Which of the following **correctly** describes the primary mode of action of an IUD, LNG-20?

(1) It suppresses sperm motility and fertilising capacity by releasing copper ions.
(2) It makes the uterus unsuitable for implantation and cervix hostile to sperms.
(3) They physically block the meeting of gametes.
(4) They permanently block gametogenesis.

147. In a population, the frequency of a recessive trait is 16%. What is the frequency of the dominant allele in the population?

(1) 0.16
(2) 0.36
(3) 0.4
(4) 0.6

148. How many of the following pairs are **mismatched**?

I. Elephantiasis - *Wuchereria malayi*
II. Amoebic dysentery - *Entamoeba histolytica*
III. Malignant malaria - *Plasmodium vivax*
IV. Ringworm - *Epidermophyton*

(1) One (2) Two
(3) Three (4) Four

149. A circular plasmid vector and a linear DNA fragment are both treated with the restriction endonuclease *Eco*RI. To create a stable recombinant DNA molecule, the sticky ends must first undergo base-pairing (annealing) before DNA ligase seals the backbone. Given that the *Eco*RI recognition sequence is 5'-GAATTC-3' and it cleaves between G and A, how many total hydrogen bonds are formed by the complementary base-pairing of the sticky ends when one DNA fragment is integrated into the vector?

(1) 12 (2) 14
(3) 16 (4) 18

150. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Introducing genetically engineered lymphocytes producing functional ADA at the early embryonic stage can permanently cure ADA Deficiency.

Reason R: Lymphocytes extracted from the blood of the patient, grown in culture and retrovirally infected with functional ADA cDNA, are immortal cells that survive indefinitely in the patient's body.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

151. The presence of a chitinous exoskeleton, and jointed appendages, are the defining characteristic of which of the following phyla?

(1) Annelida (2) Mollusca
(3) Arthropoda (4) Echinodermata

152. If the pH of the solution is equal to isoelectric point of an amino acid, what will be the net charge on the amino acid molecule?

(1) Neutral (2) Negative
(3) Positive (4) Alternating

[Contd...

153. How many of the following statements regarding animal tissues are **correct?**

A. Gap junctions facilitate the cells to communicate with each other.
B. Adhering junctions perform cementing to keep neighbouring cells together.
C. Compound epithelium is made of a single layer of tall and slender cells.
D. In all connective tissues except blood, the cells secrete fibres of structural proteins.
E. Smooth muscle fibres are fusiform, non-striated and voluntary in nature.

(1) One (2) Three
(3) Two (4) Four

154. Which of the following can recognize changes in CO_2 and H^+ concentration to send necessary signals to the respiratory rhythm centre?

(1) Chemosensitive area situated adjacent to the rhythm centre
(2) Receptors associated with the aortic arch and carotid artery
(3) Both (1) and (2)
(4) Pneumotaxic centre located in the pons region

155. If a healthy individual has 120 mL and 50 mL of blood in ventricle at the end of diastole and systole of ventricle respectively, and their heart beats 75 times per minute, what is their cardiac output?

(1) 5250 mL (2) 5000 mL
(3) 9000 mL (4) 3750 mL

156. Find the **incorrect** statements from the following regarding the concentration of filtrate in kidney.

A. The flow of filtrate in the two limbs of Henle's loop is in opposite directions.
B. The descending limb of the loop of Henle is permeable to electrolytes.
C. NaCl is transported by the ascending limb of Henle's loop which is exchanged with the descending limb of *vasa recta*.
D. Small amounts of urea enter the thin segment of the ascending limb of Henle's loop.
E. Human kidneys can produce urine nearly four times more concentrated than the initial filtrate.

(1) A and C (2) A, B and D
(3) B only (4) C only

157. Which of the following are vertebrochondral ribs?

(1) 11th and 12th pairs
(2) 6th, 7th, and 8th pairs
(3) 8th, 9th, and 10th pairs
(4) 1st to 7th pairs

158. Consider the following statements:

Statement I: The midbrain is located between the thalamus/hypothalamus of the forebrain and the pons of the hindbrain.

Statement II: Brain stem is a part of spinal cord.

In the light of the above statements, choose the most appropriate answer from the options given below

(1) Both Statement I and Statement II are correct.
(2) Both Statement I and Statement II are incorrect
(3) Statement I is correct but Statement II is incorrect.
(4) Statement I is incorrect but Statement II is correct.

159. Which of the following sets of hormones primarily interact with intracellular receptors to regulate gene expression or chromosome function?

(1) Insulin, Glucagon, Epinephrine
(2) Cortisol, Testosterone, Estradiol
(3) FSH, LH, TSH
(4) Vasopressin, Oxytocin, ANF

160. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Lack of menstruation in a 30 years old female may be indicative of pregnancy.

Reason R: During pregnancy, the persistent corpus luteum secretes large amounts of progesterone, which maintains the endometrium.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

161. According to the Medical Termination of Pregnancy (Amendment) Act, 2017, the opinion of two registered medical practitioners is required for the termination of a pregnancy if the duration of the pregnancy is:

(1) Less than 12 weeks
(2) More than 12 weeks but fewer than 24 weeks
(3) Up to the first trimester exactly
(4) More than 24 weeks but fewer than 30 weeks

[Contd...

162. Consider the following statements:
Statement I: *Homo erectus* lived about 1.5 million years ago and had a brain capacity of around 900 cc.
Statement II: The Neanderthal man with a brain size of 1400 cc lived in the East African grasslands.
In the light of the above statements, choose the most appropriate answer from the options given below
(1) Both Statement I and Statement II are correct
(2) Both Statement I and Statement II are incorrect
(3) Statement I is correct but Statement II is incorrect
(4) Statement I is incorrect but Statement II is correct

163. In which of the following diseases body attacks its own self-cells, resulting in damage?
(1) Gout
(2) Asthma
(3) Myasthenia gravis
(4) Osteoporosis

164. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Lysozyme	(I)	Degradation of fungal cell wall
(B)	Genetic engineering	(II)	Autonomously replicating circular extra-chromosomal DNA
(C)	Chitinase	(III)	Degradation of bacterial cell wall
(D)	Plasmid	(IV)	Techniques to alter the chemistry of genetic material

Choose the **correct** answer from the options given below:
(1) A-III, B-I, C-IV, D-II
(2) A-III, B-IV, C-I, D-II
(3) A-IV, B-III, C-I, D-II
(4) A-I, B-IV, C-III, D-II

165. Which of the following **accurately** describes the mechanism used to protect tobacco plants against the nematode *Meloidegyne incognita*?
(1) The introduced nematode-specific genes express crystalline insecticidal proteins.
(2) The introduced DNA produces both sense and anti-sense RNA in the host cells, which form dsRNA and silence specific nematode mRNA.
(3) The transgenic plant's roots are modified to physically block the entry of the nematode.
(4) The plant produces a viral oncogene that infects and kills the nematode upon ingestion.

166. Identify the **incorrect** statement regarding the class Osteichthyes (bony fishes):
(1) They possess four pairs of gills.
(2) Their skin is covered with cycloid scales.
(3) They possess a poison sting or electric organs.
(4) An air bladder is present.

167. In the glycogen, the right and left ends are:
(1) Reducing end and non-reducing end respectively.
(2) Non-reducing end and reducing end respectively.
(3) N-terminal and C-terminal respectively.
(4) C-terminal and N-terminal respectively.

168. In the male *Rana tigrina*, a copulatory pad is located on the:
(1) Second digit of the forelimbs
(2) Third digit of the hindlimbs
(3) First digit of the forelimbs
(4) First digit of the hindlimbs

169. If, TV = 500 mL, IRV = 3000 mL, ERV = 1100 mL, and RV = 1200 mL. What is the FRC?
(1) 2300 mL (2) 1600 mL
(3) 4600 mL (4) 5800 mL

170. How many of the following are **correctly** matched.
I. Neutrophils - Phagocytic cells that destroy foreign organisms
II. Basophils - Secrete histamine, serotonin and are involved in inflammatory reactions
III. Eosinophils - Associated with allergic reactions and resist infections
IV. B-lymphocytes - Primarily responsible for cell-mediated immunity
(1) One (2) Two
(3) Three (4) Four

171. Match **List-I** with **List-II**.

List-I		List-II	
(A)	Hyoid	(I)	Acts as fulcrum
(B)	Joints	(II)	U-shaped bone
(C)	Addison's disease	(III)	Allows passage of small amounts of urea
(D)	Collecting Duct	(IV)	Hyposecretion of hormones of adrenal cortex

Choose the **correct** answer from the options given below:
(1) A-II, B-I, C-III, D-IV
(2) A-II, B-IV, C-I, D-III
(3) A-III, B-I, C-IV, D-II
(4) A-II, B-I, C-IV, D-III

[Contd...

172. Read the following statements and choose the option which states them as True or False **correctly**.

I. In frogs, gall bladder secretes bile that is stored in the liver.

II. Cardiac muscle cells possess intercalated discs which allow them to contract together as a unit.

III. In some countries the muscular legs of frog are used as food by man.

IV. The A-band of a sarcomere contains only thin filaments.

(1) F, T, T, F
(2) T, T, F, F
(3) F, F, T, T
(4) F, T, F, T

173. Which of the following is **not** primarily regulated by the limbic system?

(1) Regulation of sexual behaviour
(2) Expression of emotional reactions like excitement and rage
(3) Cardiovascular reflexes and gastric secretions
(4) Motivation

174. Which of the following serves as a classic example of convergent evolution?

(1) The forelimbs of whales, bats, cheetahs, and humans
(2) The vertebrate hearts and brains
(3) The thorns of *Bougainvillea* and the tendrils of *Cucurbita*
(4) The eye of the octopus and the eye of mammals

175. What is added at the end of DNA isolation, to precipitate purified DNA?

(1) Ribonuclease
(2) Chilled ethanol
(3) Ethidium bromide
(4) Agarose gel

176. Which ART is employed, if the embryo formed *In vitro* has more than 8 blastomeres?

(1) ZIFT
(2) GIFT
(3) IUT
(4) AI

177. Which of the following represents the **correct** evolutionary lineage leading towards Angiosperms?

(1) Chlorophyte ancestors → Tracheophyte ancestors → Rhynia-type plants → Psilophyton → Seed ferns → Angiosperms
(2) Tracheophyte ancestors → Chlorophyte ancestors → Psilophyton → Rhynia-type plants
(3) Rhynia-type plants → Chlorophyte ancestors → Psilophyton → Progymnosperms
(4) Psilophyton → Tracheophyte ancestors → Seed ferns → Conifers

178. Consider the following statements:

Statement I: Benign tumors normally remain confined to their original location.

Statement II: Malignant tumors consist of neoplastic cells that grow rapidly and heavily rely on a property called contact inhibition to limit their spread.

In the light of the above statements, choose the most appropriate answer from the options given below

(1) Both Statement I and Statement II are correct
(2) Both Statement I and Statement II are incorrect
(3) Statement I is correct but Statement II is incorrect
(4) Statement I is incorrect but Statement II is correct

179. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: If an alien DNA is ligated specifically at the *BamH* I site, the recombinant plasmid will lose its resistance to ampicillin.

Reason R: The recognition site for the restriction enzyme *BamH* I is located within the tetracycline resistance gene, leading to insertional inactivation.

In the light of the above statements, choose the **correct** answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A
(2) Both A and R are true but R is not the correct explanation of A
(3) A is true but R is false
(4) A is false but R is true

180. In what year was Rosie produced and what was the concentration of human protein present in its milk?

(1) 1990, 2.4 g/litre (2) 1997, 2.4 g/litre
(3) 1997, 4.2 g/litre (4) 1983, 2.4 g/litre

■■■

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